

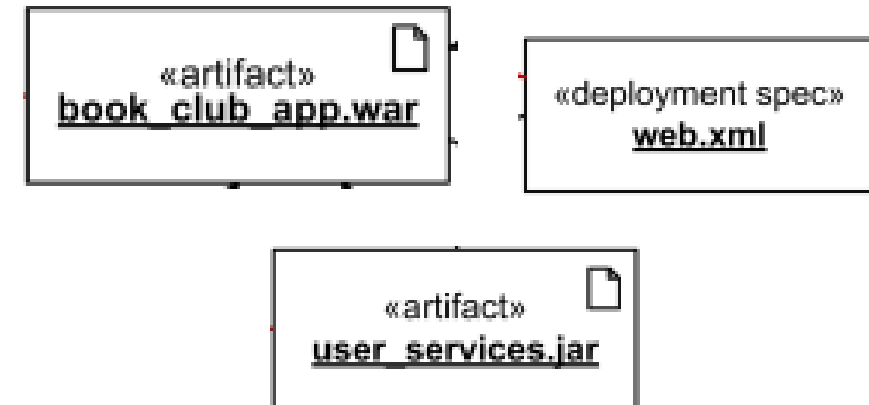
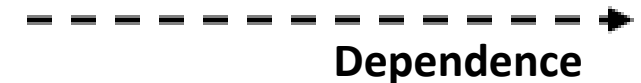
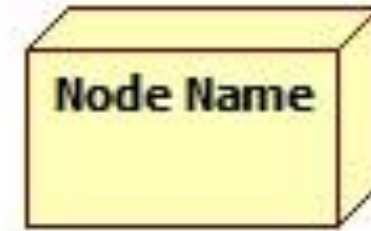
Architectural design Guide

4/2019

Notes for Proj#3

Deployment view

- **Artifacts** represent concrete elements in the physical world that are the result of a development process.
 - Examples of artifacts are **executable files, libraries, archives, database schemas, configuration files, etc.**
- **Deployment target** is usually represented by a **node** which is either **hardware device** or **some software execution environment**.
 - Nodes could be connected through **communication paths** to create networked systems of arbitrary complexity.



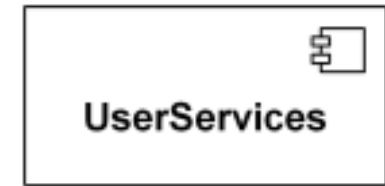
Deployment view

- A **component** is a **class** representing a modular part of a system with encapsulated content and whose **manifestation** is replaceable within its environment.
- A component has its behavior defined in terms of **provided interfaces** and **required interfaces** (potentially exposed via **ports**).

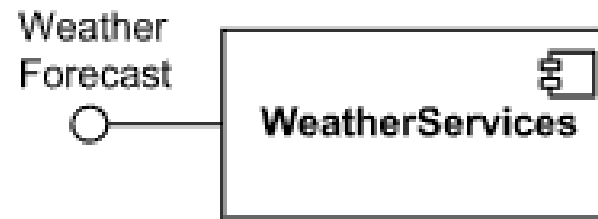
in UML 1.4



*WeatherServices
component*



*UserServices
component*



provided interface

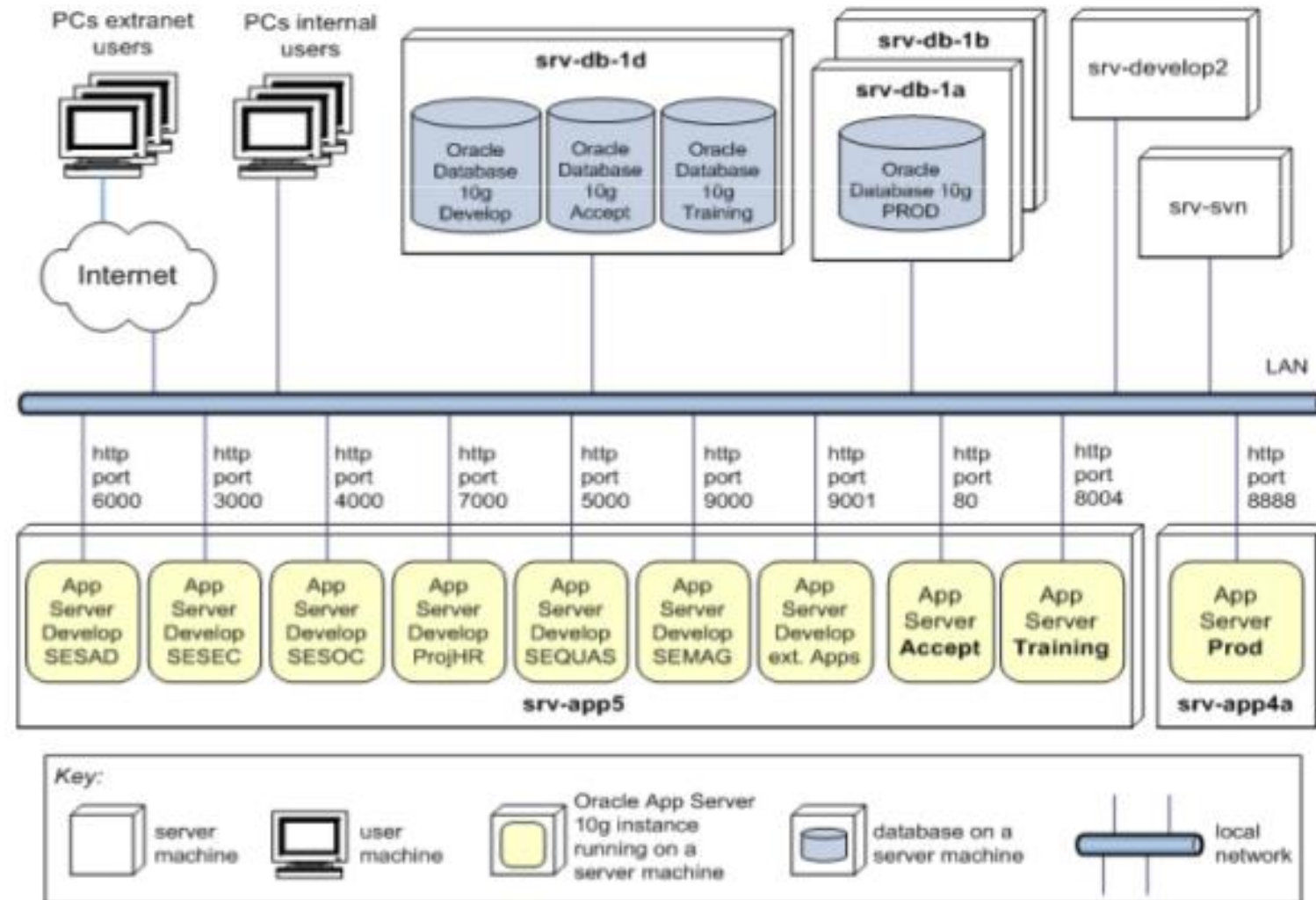


required interface

<https://sites.google.com/site/softwarearchitectureinpractice/9-documenting-software-architecture/d-allocation-views/a-deployment-view>

Deployment view

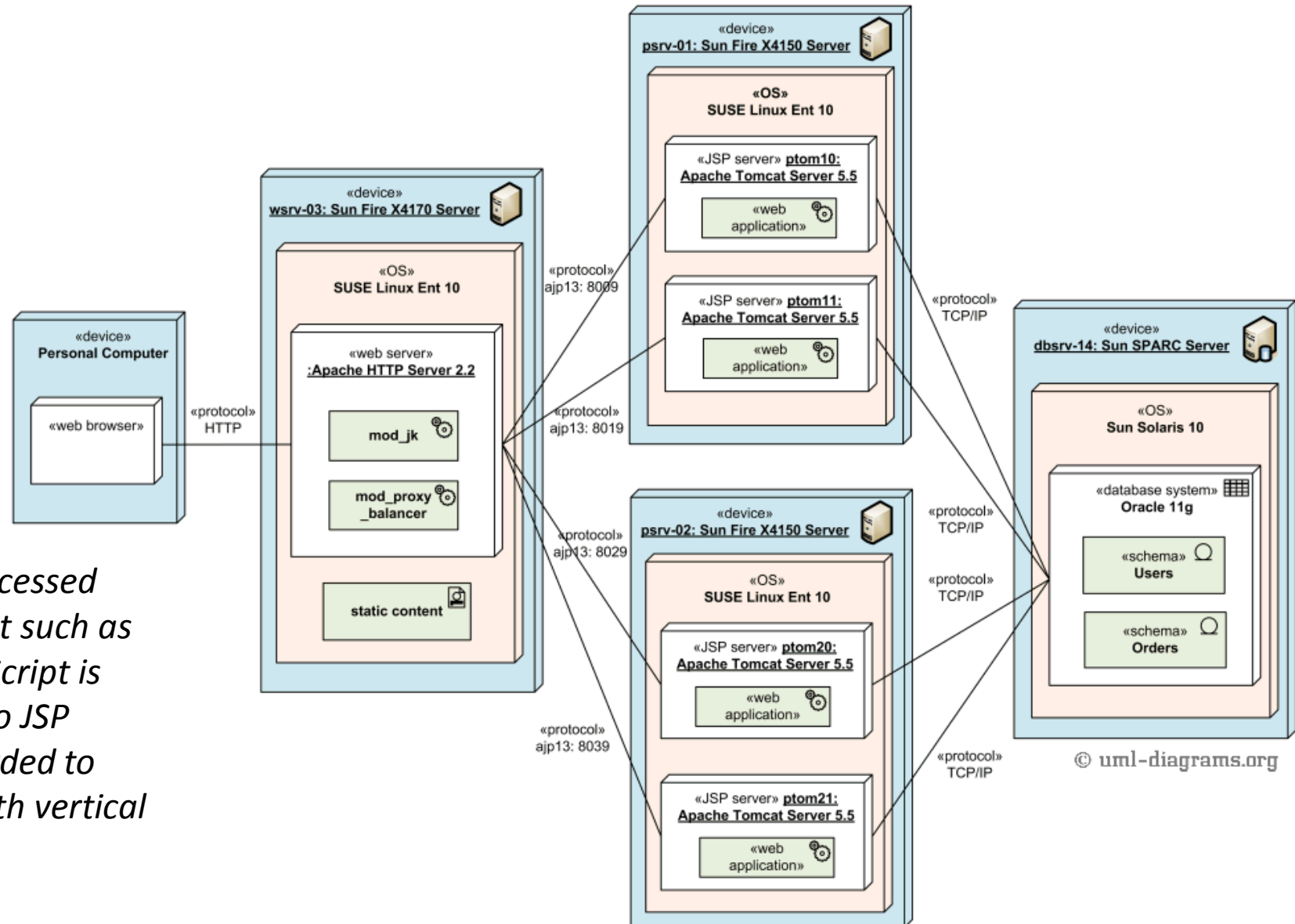
- Good example



Deployment view

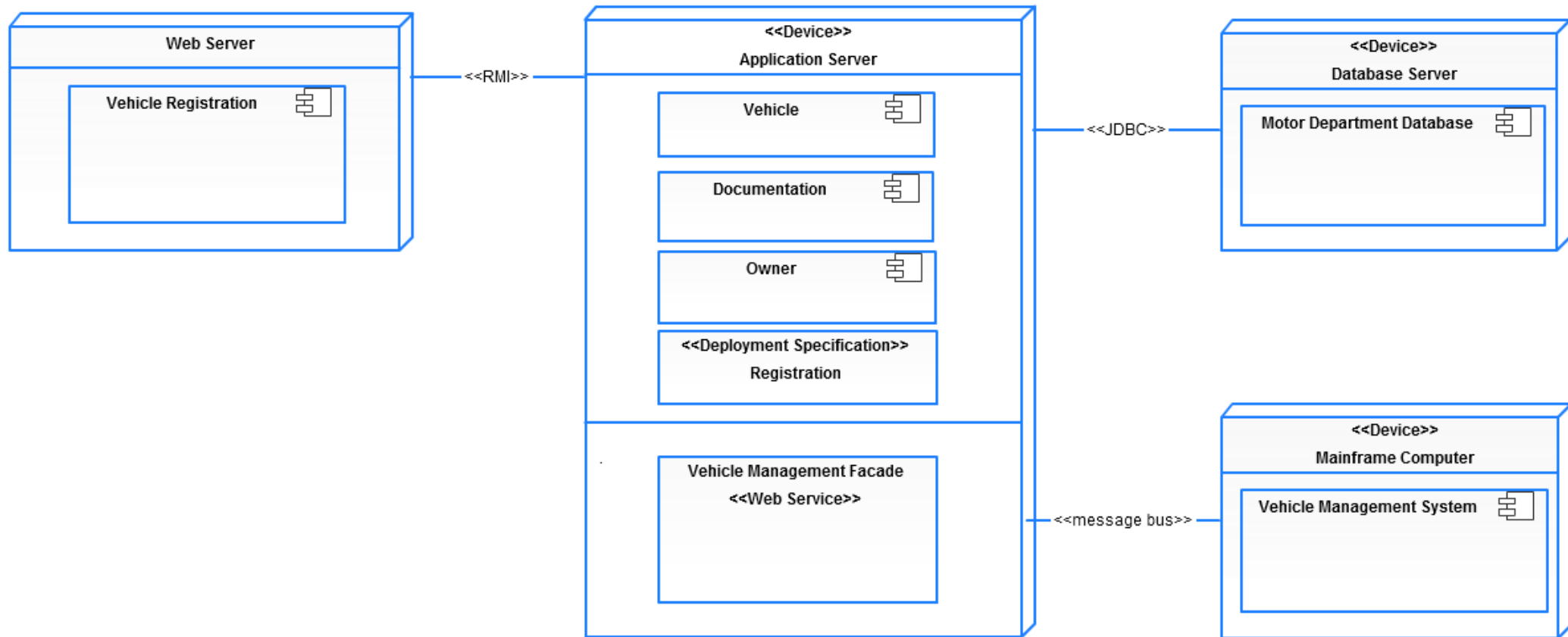
An example of **deployment diagram** for J2EE web application with load balancing and clustering which shows specific server instances involved.

Incoming HTTP requests are first processed by **Apache web server**. Static content such as HTML pages, images, CSS, and JavaScript is served by the web server. Requests to JSP pages are **load balanced** and forwarded to 2x2 Apache Tomcat servers using both vertical and horizontal **clustering**.

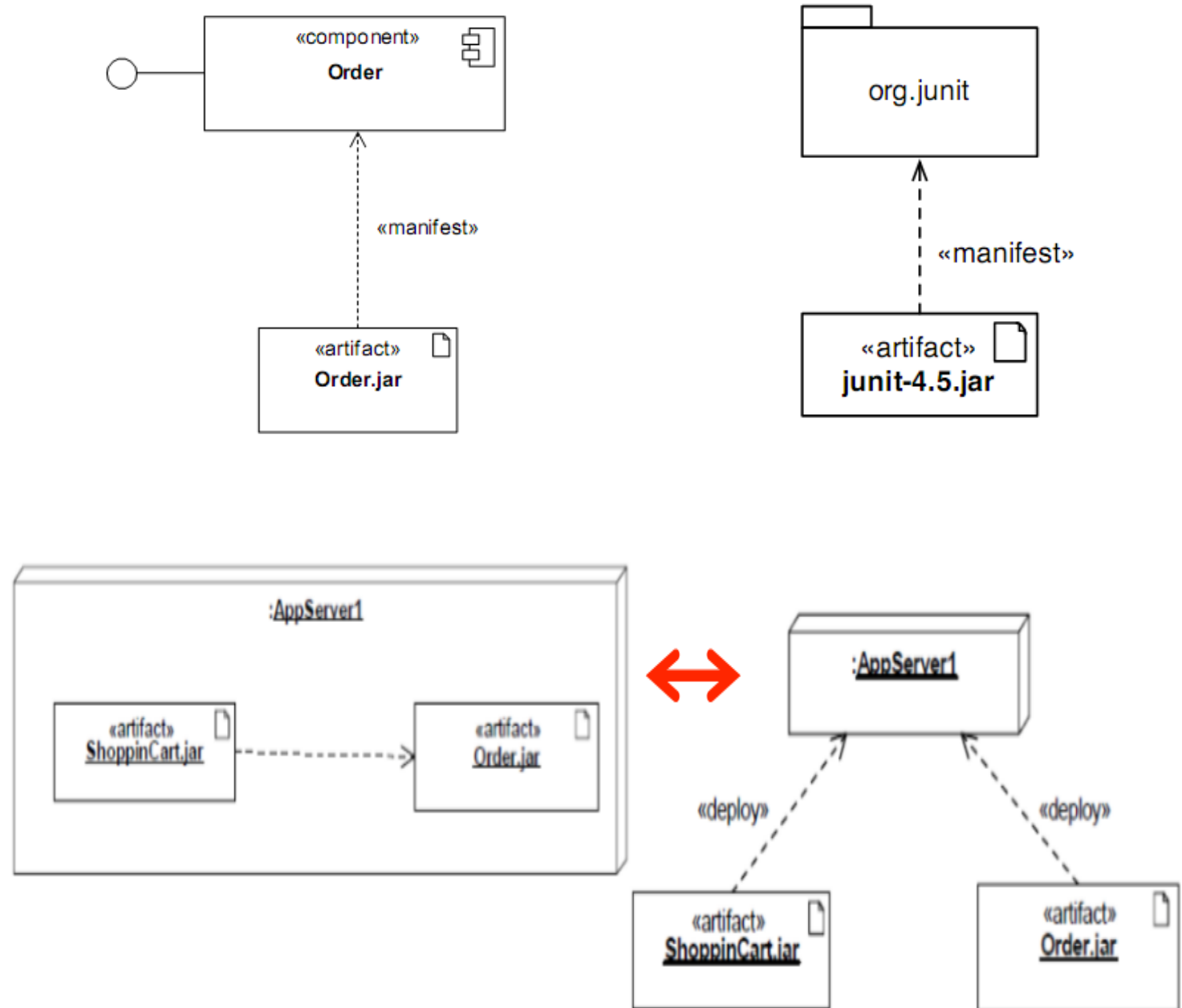


Deployment view

Deployment Diagram For a Vehicle Registration System

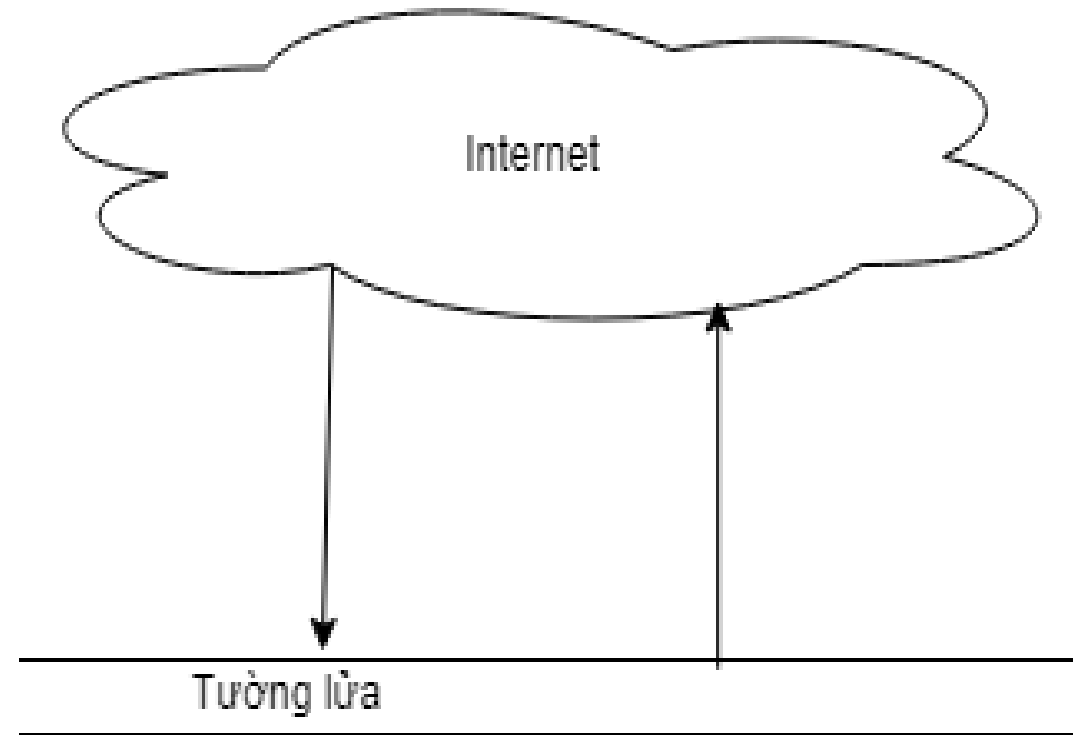


Note, that components were directly deployed to nodes in UML 1.x deployment diagrams. In UML 2.x artifacts are deployed to nodes, and artifacts could manifest (implement) components. Components are deployed to nodes indirectly through artifacts.



Deployment view

- ???

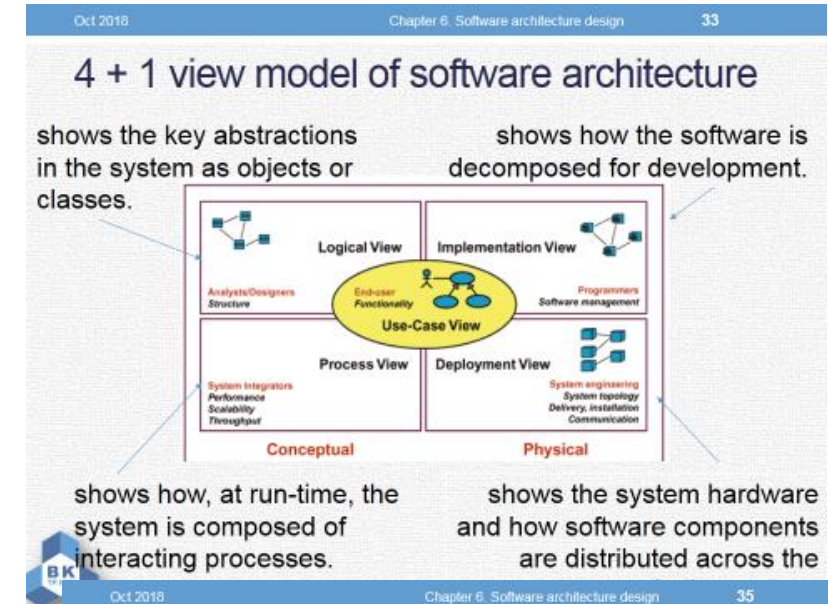


Deployment view

- Cần rõ các <<subsystem>>/component và vị trí vật lý (cài đặt)
- Hoặc chỉ rõ thành phần nào sẽ deploy lên server, client, ...

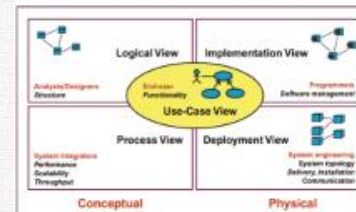
Development/Implementation view

- Component diagram / Package diagram
- Nên dùng các architectural design pattern
 - Layered, Repository, Client-Server, ...
- Các thành phần bên trong component/package
 - Các component/package
 - Không là function/method

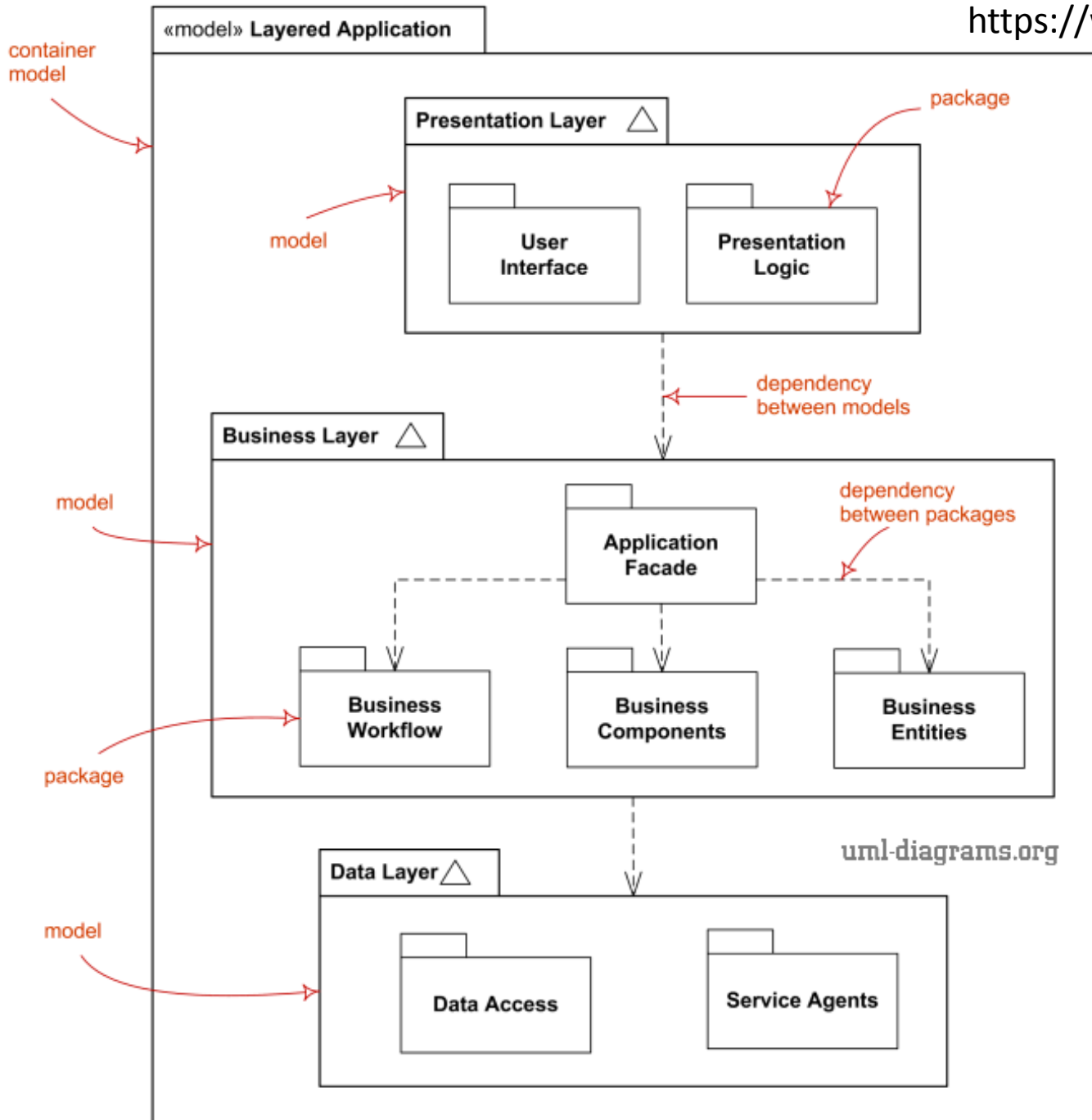


4 + 1 view model vs UML

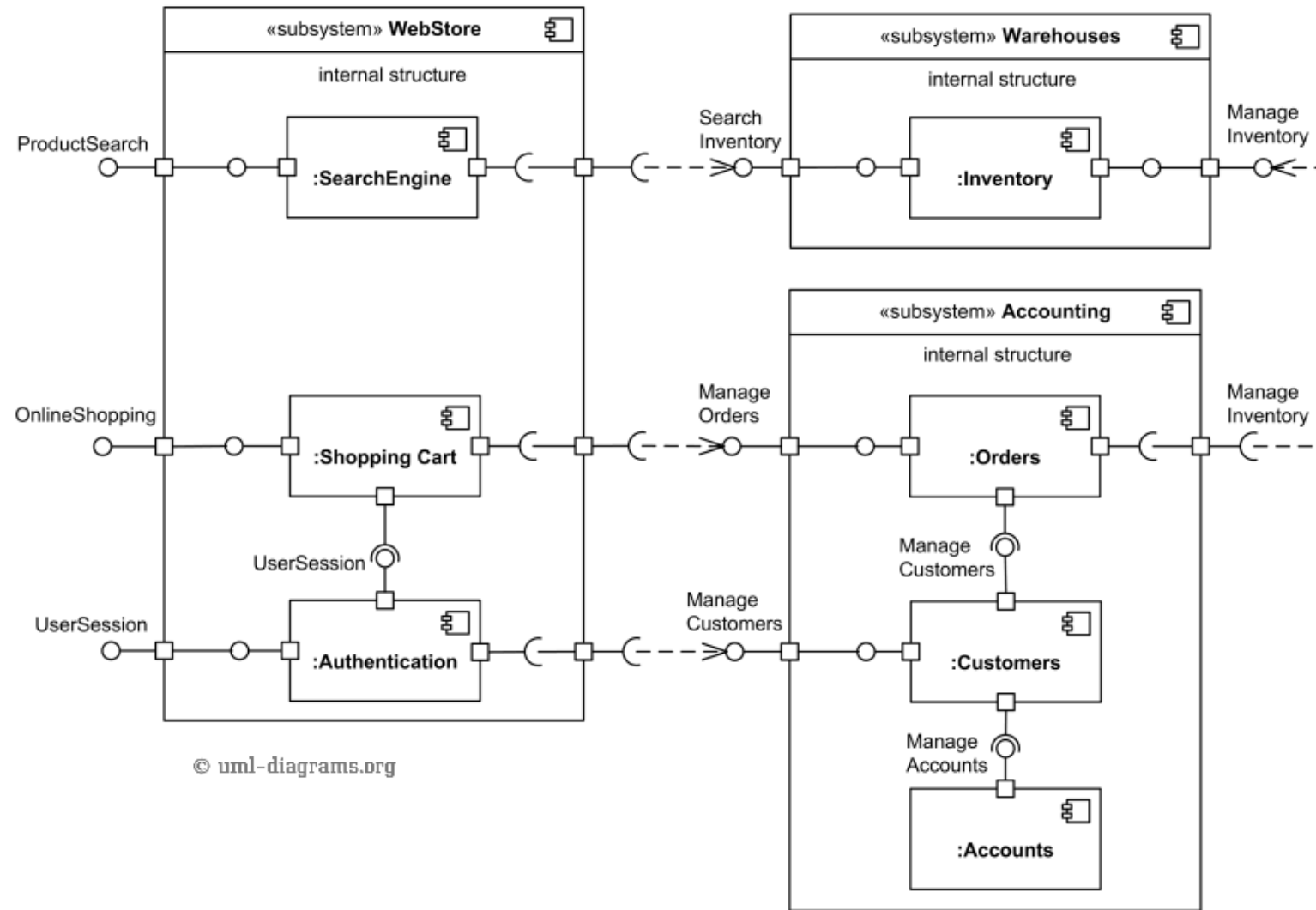
- Logical view:
 - Class diagram, Communication diagram, Sequence diagram
- Process view:
 - Activity diagram
- Development view:
 - Component diagram, Package diagram.
- Physical view:
 - Deployment diagram
- Scenarios (+1):
 - Use-case



Development/Implementation view



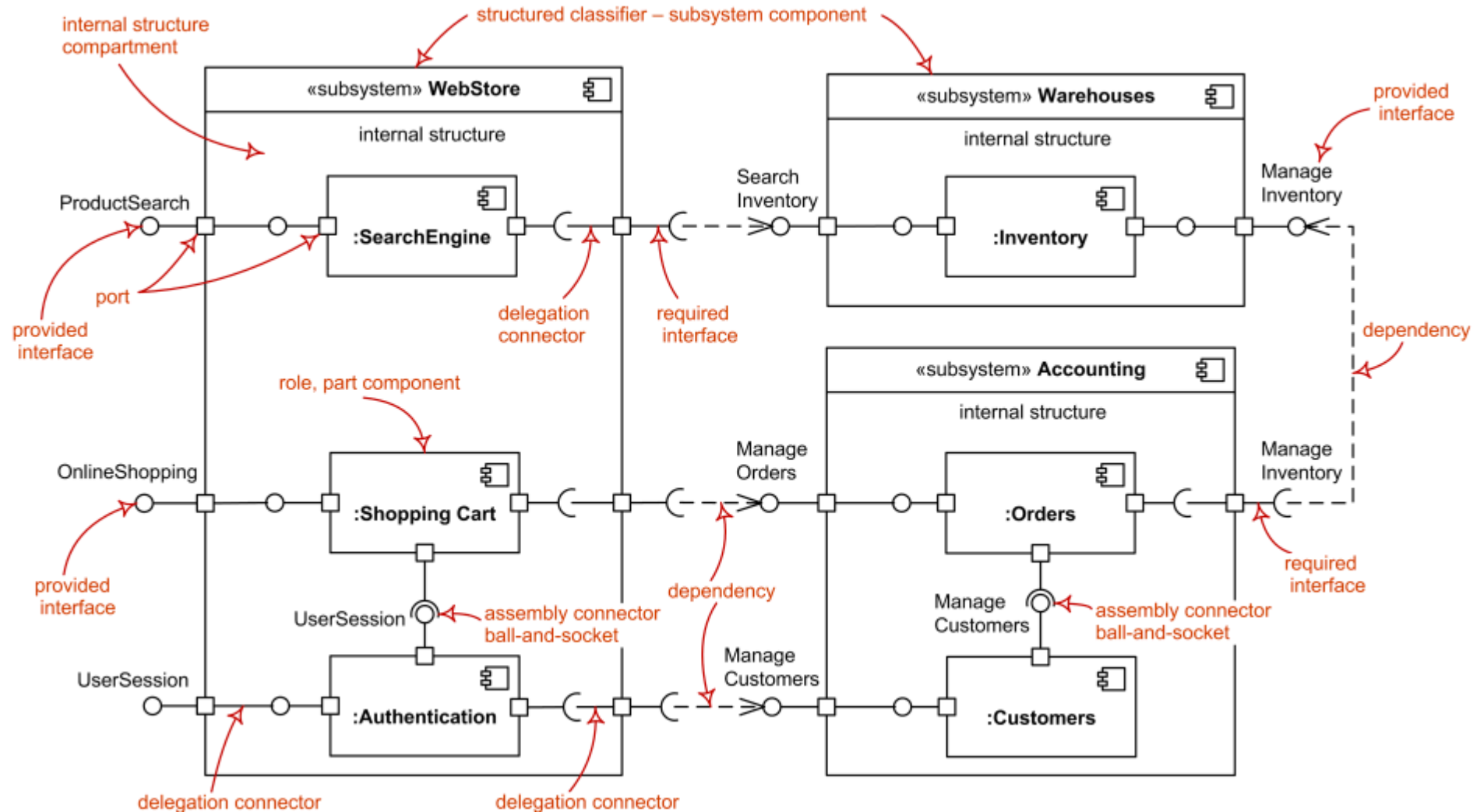
- Online shopping component diagram
 - → internal structure of three related subsystems- WebStore, Warehouses, and Accounting



Desc

- **WebStore** subsystem contains three components related to online shopping - **Search Engine**, **Shopping Cart**, and **Authentication**.
 - **Search Engine** component allows to search or browse items by exposing provided interface **Product Search** and uses required interface **Search Inventory** provided by **Inventory** component.
 - **Shopping Cart** component uses **Manage Orders** interface provided by **Orders** component during checkout.
 - **Authentication** component allows customers to create account, login, or logout and binds customer to some account.
- **Accounting** subsystem provides two interfaces - **Manage Orders** and **Manage Customers**. Delegation connectors link these external contracts of the subsystem to the realization of the contracts by **Orders** and **Customers** components.
- **Warehouses** subsystem provides two interfaces **Search Inventory** and **Manage Inventory** used by other subsystems and wired through dependencies.

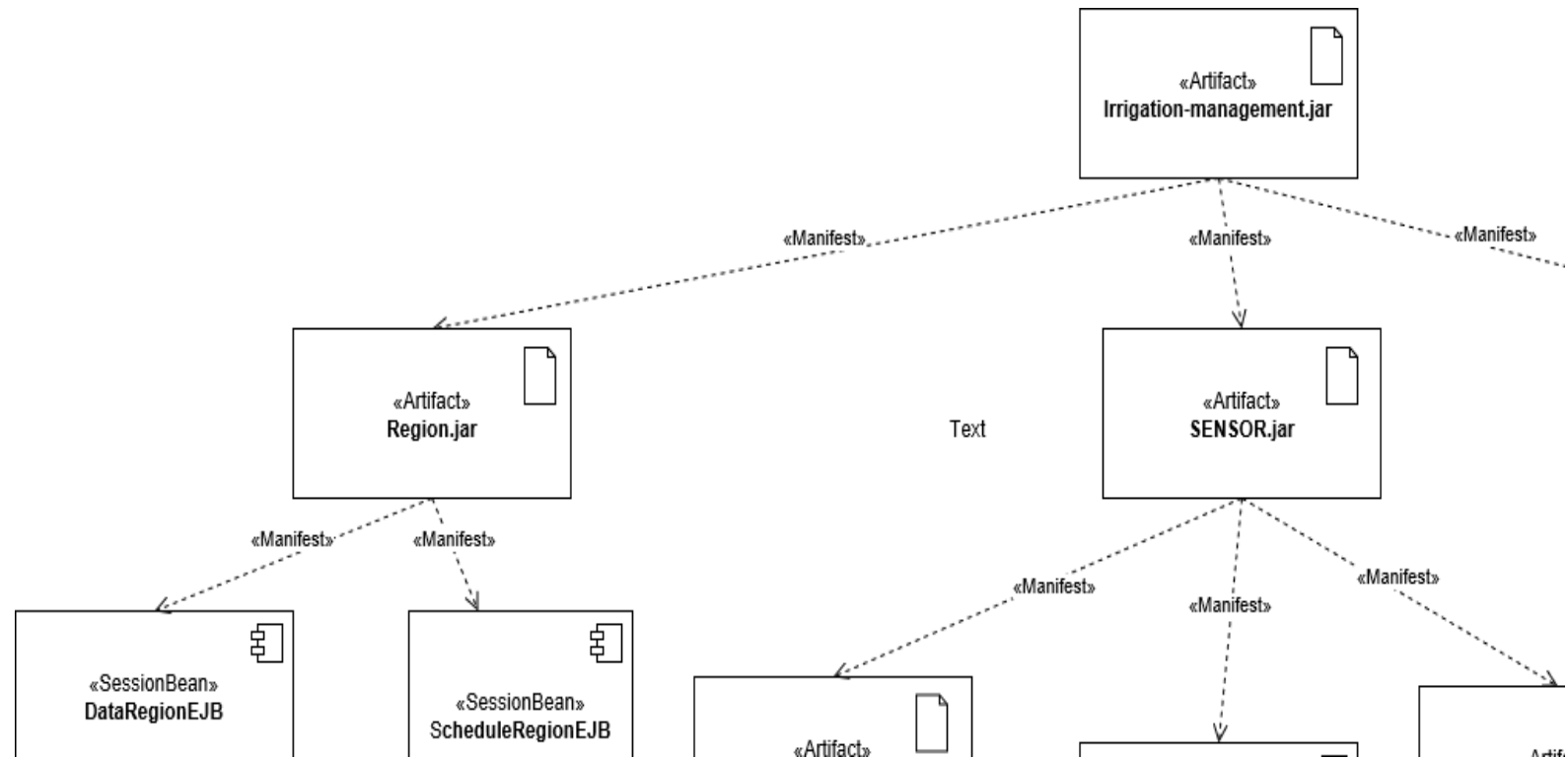
Development/Implementation view



- The major elements of UML component diagram - component, provided interface, required interface, port, connectors.

Development/Implementation view

- .jar files inside .jar file, possible?



Submit for Project #3

- Deployment view (general architecture): 5
 - Group work
- Implementation view (module view, programming view): 5
 - Individual work

	Quá dưới mức yêu cầu (0% đ)	Dưới mức yêu cầu (50% đ)	Gần đạt yêu cầu (75% đ)	Đạt yêu cầu (100% đ)	Bonus	Điểm
	No or almost no	Under expectation	Almost at the expectation	Expectation		Point
Deployment view	Không có	Sơ sài, chưa rõ kiến trúc tổng quan hệ thống	Vẫn còn đơn giản	Đầy đủ, rõ ràng		5
	No		Simple	Good		
		Simple, not a clear general architecture (deployment view)				
	0	2.5	4	5		
Implementation view	Không có	Sơ sài, chưa rõ các khối hiện thực	Vẫn còn đơn giản	Đầy đủ, rõ ràng		5
	No		Simple	Good		
		Simple, not a clear implementation view				
	0	2.5	4	5		